Software Requirements Specification

EduBridge — AI-Powered Education Management & Learning Intelligence SaaS

Prepared for: Private Educational Institutions (Schools, Academies, Tutorial Centers), with a future expansion path toward Regional and National Education Systems

Version 2.0 — Reshaped from the original national-infrastructure SRS — August 2026

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1. Introduction

1.1 Purpose

This SRS specifies requirements for the reshaped EduBridge: an AI-powered education management and learning-intelligence SaaS platform serving private educational institutions (schools, academies, tutorial centers). It replaces the original Ministry-dependent, national-infrastructure scope with an institution-first product, delivered in three incremental versions.

1.2 Scope

EduBridge v1 will deliver institution/academic management with hierarchical account provisioning (Institution Administrator → Teachers → Students), dynamic role permissions resolved via lookup rather than hardcoded checks, curriculum-aligned content and learning materials, and AI-assisted content generation for both Grade 8/12 exam preparation and general competency-based assessments — all subject to mandatory teacher review and approval before publication to students. Students receive a system-generated default password at account creation. Offline-first access and one-way SMS progress notifications to parents/guardians are introduced in v2. The conversational AI Tutor and classical machine-learning-based prediction (at-risk detection, mastery/engagement forecasting) are introduced in v3, once the platform has real usage data and a proven content base to ground them on. Government/Ministry-scale integration remains a future expansion beyond v3, not an initial dependency.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SaaS	Software as a Service, multi-tenant institution subscription model
Tenant	A single subscribing institution and its isolated data
AI Tutor	Conversational AI assistant that explains concepts and guides students (v3)
Exam Prep Module	AI-assisted practice environment built on teacher-authored question banks for Grade 8/12 regional & national exams
Mastery	Estimated competency level of a student on a curriculum topic
Classical ML	Non-deep-learning models (e.g., logistic regression, gradient boosting, decision trees) used for prediction
Dynamic Permission Lookup	A role × action lookup table the system consults at request time to resolve access, instead of hardcoded role checks
Default Password	A system-generated, single-use password issued only to Student accounts at creation, requiring a forced change on first login

1.4 References

- IEEE Std 830-1998 (SRS structure)
- Original EduBridge national-infrastructure SRS, v1.0 (superseded scope)
- EduBridge Reshaping Proposal — Business-Focused Education Platform
- National curriculum frameworks for Grade 8 and Grade 12 examinations

1.5 Document Overview

Section 2 gives the overall product description. Section 3 details functional requirements, grouped by release version. Section 4 covers interfaces. Section 5 covers non-functional requirements. Section 6 defines the data model. Section 7 describes architecture. Section 8 details core algorithms. Section 9 contains appendices.

2. Overall Description

2.1 Product Perspective

EduBridge is a new multi-tenant B2B SaaS product. Each institution is an isolated tenant. The platform is built and demonstrated independently of any government system; Ministry/regional integration is a later expansion, not a dependency.

2.2 Product Functions Summary (by version)

Version	Core Functions
v1	Institution/academic management · Hierarchical account provisioning (Admin → Teacher → Student) with system-generated default passwords for Students · Dynamic permissions via lookup · Curriculum-aligned content & learning materials · Competency assessment · Teacher analytics · AI-assisted content generation for Grade 8 & 12 exam prep and general competency assessments (teacher-approved) · Multi-tenancy
v2	Offline-first access (caching + sync) · SMS/USSD progress notifications to parents/guardians (no portal)
v3	AI Tutor (conversational, RAG-grounded) · Classical ML at-risk prediction · Mastery/engagement forecasting · Advanced analytics

2.3 User Classes and Characteristics

User Class	Description	Primary Device
Student (Grade 8 / Grade 12 focus for exam prep)	Consumes content, practices exam questions, completes assessments; logs in with a system-generated default password on first use	Mobile/Web PWA
Teacher	Manages classes, authors exam question banks, reviews analytics, approves AI-generated content	Web Dashboard
School Administrator	Manages institution setup, enrollment, subscription, creates Teacher accounts	Web Dashboard
Parent/Guardian (v2+)	Receives periodic one-way SMS progress notifications; no login and no portal	SMS

2.4 Operating Environment

- Student/Teacher apps: responsive web app (Next.js), mobile-friendly PWA.
- v2 offline mode: service-worker caching for low-connectivity institutions.
- v2 SMS: integration with a mobile network operator gateway for one-way parent notifications.

2.5 Design and Implementation Constraints

- Multi-tenant data isolation is mandatory from v1 (shared schema + tenant_id enforcement).
- Access control is resolved dynamically via a permissions lookup (role × action × tenant override), not hardcoded per role, so institutions can customize access without a code change.
- AI-generated exam-prep and competency-assessment content must be traceable to teacher-authored source questions/materials and always requires teacher review/approval before publication; AI does not invent or publish content unsupervised.
- Classical ML models (v3) must expose the top contributing factors behind any risk score — no black-box-only output.

2.6 Assumptions and Dependencies

- Institutions provide their own teacher rosters at onboarding; Teacher accounts are created by the Institution Administrator, and Student accounts by an Administrator or Teacher.
- Teachers are responsible for authoring/curating the Grade 8 & 12 exam question bank and for reviewing/approving any AI-generated content before it reaches students.
- v2 SMS notifications require a contracted telecom gateway per operating region.

3. System Features and Functional Requirements

3.1 Version 1 — Core Platform

FR-01 — Institution & Academic Management (Critical)

FR-01.1: Manage students, teachers, classes, subjects, academic years, sections, enrollments. FR-01.2: Access to every action is resolved dynamically through a permissions lookup (role × action × tenant-level override), rather than a hardcoded per-role check, so an institution's permission set can be adjusted without a code change. FR-01.3: Full tenant data isolation across all modules.

FR-02 — Hierarchical Account Provisioning (Critical)

FR-02.1: Onboarding follows a strict top-down order: the Institution Administrator account is created first (at subscription sign-up), before any Teacher or Student account can exist for that tenant. FR-02.2: Only an authenticated Institution Administrator can create Teacher accounts for their tenant. FR-02.3: Only an authenticated Administrator or Teacher can create Student accounts, and each Student is linked to the creating institution/class at creation time. FR-02.4: Every newly provisioned Student account receives a system-generated default password, single-use and issued to the enrolling Administrator/Teacher for hand-off to the student. FR-02.5: A Student logging in with the default password is forced to set a new password before accessing any other part of the platform. FR-02.6: Default passwords expire after a configurable period if unused; an Administrator or Teacher can regenerate credentials for a locked-out Student account. FR-02.7: All account-creation and credential-reset actions are logged for audit, tagged with the creating user's role.

FR-03 — Curriculum-Aligned Content & Learning Materials (Critical)

FR-03.1: Content library covers each institution's subjects/grades, with each item tagged to a specific curriculum topic or learning objective. FR-03.2: Teachers create, publish, and version lessons, learning materials, assignments, quizzes, and homework; students access assigned materials and submit work through the platform. FR-03.3: Content supports multiple formats where feasible (text, audio, video/illustration) to support diverse learning needs.

FR-04 — AI-Assisted Exam Preparation & Competency Content Generation (Critical)

FR-04.1: Teachers author and maintain the official question bank per subject, aligned to Grade 8 and Grade 12 regional/national exam formats. FR-04.2: Students practice using teacher-authored questions in timed/untimed exam-simulation mode. FR-04.3: AI assists students during Grade 8/12 exam practice by explaining wrong answers, generating similar practice items from the teacher's question bank, and highlighting weak topics — the AI does not replace or auto-publish new official exam questions without teacher review. FR-04.4: Exam-prep performance feeds into the student's mastery/topic breakdown (see FR-05). FR-04.5: Teachers see class-wide exam-readiness analytics per subject/topic ahead of Grade 8/12 exam dates. FR-04.6: Beyond Grade 8/12 exam prep, AI can generate draft competency-based assessment questions and content for any grade/subject the institution teaches; every AI-generated item is queued for mandatory teacher review and explicit approval, and is never visible to students until approved.

FR-05 — Competency-Based Assessment & Student Intelligence (High)

FR-05.1: Quizzes/exams/exam-prep attempts update a per-topic mastery status (Not Started / Developing / Proficient / Mastered). FR-05.2: System surfaces each student's weak topics, strong topics, and performance trend.

FR-06 — Teacher Intelligence Dashboard (High)

FR-06.1: Class and per-student performance analytics, including exam-readiness view for Grade 8/12 classes. FR-06.2: Weak-topic identification with suggested interventions.

3.2 Version 2 — Offline Access & SMS Notifications

FR-07 — Offline-First Access (Critical for v2)

FR-07.1: Core learning content and exam-prep question sets are available offline once downloaded/cached. FR-07.2: Assessment/practice attempts taken offline queue locally and sync automatically when connectivity returns, with conflict resolution favoring the most complete record. FR-07.3: Institutions with low connectivity can pre-load content bundles for a class/term. FR-07.4: A user's first-login forced password reset (FR-02.5) is also enforceable offline, syncing the new credential once connectivity returns.

FR-08 — Parent/Guardian SMS Notifications (High for v2)

FR-08.1: Periodic SMS progress summaries are sent to each student's registered parent/guardian phone number, in the institution's local language. FR-08.2: SMS is the sole parent-facing channel — no parent login, no web portal, and no USSD menu are provided. FR-08.3: Failed SMS deliveries are retried and logged for the institution administrator to review.

3.3 Version 3 — AI Tutor & Predictive Intelligence

FR-09 — AI Tutor (Critical for v3)

FR-09.1: Students can ask the AI Tutor to explain concepts, answer questions, and give step-by-step guidance, grounded in the institution's own content library and exam question bank (RAG), built up over v1/v2 usage. FR-09.2: AI Tutor can generate additional practice questions and homework help on demand, on top of the existing curriculum content. FR-09.3: All AI Tutor sessions are logged per student for teacher visibility. FR-09.4: AI-assisted content-creation tools for teachers (draft lessons, practice sets) are introduced alongside the AI Tutor, subject to teacher approval before publishing.

FR-10 — Classical ML At-Risk & Performance Prediction (High for v3)

FR-10.1: A classical ML model (e.g., logistic regression / gradient-boosted trees) scores each student's risk of falling behind, using engagement, mastery-trend, and exam-prep performance features.FR-10.2: Risk flags include the top contributing factors for teacher-led intervention; the model never takes autonomous action on a student account. FR-10.3: Model performance (precision/recall against actual outcomes) is reviewed each term and reported to institution

admins. FR-10.4: Predictive exam-readiness scoring for Grade 8/12 students, estimating likelihood of meeting a target competency level before the exam date.

4. External Interface Requirements

4.1 User Interfaces

Student app: learning content, exam-prep practice, assignments, progress view; AI Tutor is added in v3. Teacher dashboard: class management, question-bank authoring, AI-generated content review/approval, analytics. Admin console: institution setup, subscription/billing, tenant configuration, Teacher account creation. Parent channel (v2): outbound SMS notifications only — no login, no portal.

4.2 Hardware Interfaces

Standard smartphones, tablets, and desktop browsers. No proprietary hardware required. v2 offline mode targets low-spec Android devices for institutions with limited connectivity.

4.3 Software Interfaces

Interface	Direction	Data	Version
AI/LLM Provider API	Outbound	Exam-prep assistance, competency content generation, AI Tutor (v3)	v1 / v3
SMS Gateway	Outbound	Parent progress notifications	v2
Future: Ministry Curriculum / Exam Systems	Inbound (future)	Official exam alignment reference data	Post-v3 expansion

4.4 Communications Interfaces

TLS 1.2+ for all traffic. v2 offline content bundles are integrity-checked on sync to prevent tampering.

5. Non-Functional Requirements

5.1 Performance

- Cached/loaded content: < 1 second.

- Adaptive/exam-prep question scoring: < 200 ms.

5.2 Availability & Reliability

- 99.9% uptime target for online components (v1).
- v2: offline functionality remains available regardless of backend status; sync conflicts resolved without data loss.

5.3 Scalability

Multi-tenant architecture designed to scale from a handful of pilot institutions to hundreds of institutions without redesign; tenant isolation enforced at the data layer.

5.4 Security & Privacy

- Access control enforced per tenant and per role via the dynamic permissions lookup (FR-01.2).
- Student data protected to a COPPA-equivalent standard given the presence of minors, especially Grade 8 students.
- Default passwords (Student accounts only) are single-use, transmitted once, and never stored or logged in plaintext.
- All AI-generated content, question-bank changes, and account-provisioning actions are logged for audit.

5.5 Accessibility

WCAG 2.1 AA targeted for student and teacher interfaces; full compliance audit scheduled once core v1 features stabilize.

5.6 Compliance

Data handling aligned with applicable national data-protection requirements, with heightened care for minor students' academic and engagement data.

6. Data Requirements (High-Level Data Model)

Core Entities:

Institution (Tenant), UserAccount (Administrator/Teacher/Student, with role), PermissionLookup (role × action × tenant-override), CredentialResetLog, Student, Teacher, Class, Subject, AcademicYear, ContentItem, Assignment, ExamQuestionBank, ExamPrepAttempt, Assessment, MasteryRecord, ParentGuardian (v2, phone number only — no portal login), AITutorSession (v3), RiskScore (v3).

Key relationships:

- A UserAccount is created by exactly one higher-role account (Administrator creates Teacher; Administrator or Teacher creates Student).
- Only Student accounts carry a provisioning state (Pending First Login → Active) tied to the default-password flow; Teacher/Administrator accounts are provisioned by a separate mechanism outside this scope.
- Every access check resolves against PermissionLookup for the acting role, the requested action, and any tenant-specific override.
- A Teacher authors ExamQuestionBank items tagged to Grade 8/12 subject and exam topic, and reviews/approves AI-generated competency-assessment items before they publish.
- A Student's ExamPrepAttempt and Assessment records update MasteryRecord per topic.
- (v3) AITutorSession references the Student and the ContentItem(s)/ExamQuestionBank items used to ground its answers.
- (v3) RiskScore references a Student and the contributing MasteryRecord/engagement features used to generate it.

7. System Architecture Overview

- Presentation Layer: Student/Teacher/Admin web apps (Next.js); offline-capable PWA shell added in v2.
- API Layer: REST API (NestJS) with tenant-isolation middleware, a dynamic Permissions Lookup Service (resolves role × action × tenant-override on every request), and a

Provisioning/Auth Service (Student default-password issuance and forced-reset enforcement).

- Application Services (v1): Academic Service, Provisioning/Auth Service, Permissions Lookup Service, Content Service, Exam-Prep & Competency Content Service (with AI assistance and teacher-approval workflow), Analytics Service.
- Application Services (v2): Sync Service, SMS Notification Service.
- Application Services (v3): AI Tutor Service (RAG), Prediction Service (classical ML).
- Data Layer: PostgreSQL (+pgvector, provisioned ahead of v3 for AI retrieval) via Prisma, object storage for materials, offline sync-queue store (v2).
- AI/ML Layer: LLM provider abstraction used for exam-prep and competency content generation from v1, extended to the full AI Tutor in v3; classical ML models for risk/readiness prediction added in v3.

8. Core Algorithm Specifications

8.1 Dynamic Permission Lookup (v1)

Each incoming request is checked against a PermissionLookup table keyed by (role, action, tenant). The lookup returns allow/deny before the request reaches business logic, so an institution's permission set can be adjusted by editing lookup rows rather than redeploying code.

8.2 Student Account Provisioning (v1)

On Student account creation only, the system generates a random, single-use default password, hashes it for storage, and issues the plaintext once to the creating Administrator/Teacher for hand-off. The Student account is flagged Pending First Login; every endpoint except the password-change endpoint is blocked until the password is changed, at which point the account becomes Active. Account creation itself is only permitted top-down — Administrator → Teacher → Student — enforced via the Permissions Lookup Service (8.1).

8.3 Exam-Prep & Competency Content AI Assistance (v1)

Retrieval-augmented generation constrained to each institution's own teacher-authored question bank and curriculum content. The AI grounds explanations and generated practice or assessment items in that source content and cites the originating question/topic; every generated item is queued for mandatory teacher review and is never visible to students until explicitly approved.

8.4 Competency/Mastery Tracking (v1)

Per-topic mastery status is updated from quiz, assignment, and exam-prep attempt results, feeding the student intelligence and teacher-dashboard views.

8.5 Offline Sync (v2)

Local-first storage with a background sync queue; conflict resolution favors the most complete/most recent authoritative record on reconnect.

8.6 AI Tutor (v3)

Retrieval-augmented generation over the institution's accumulated content library and exam question bank, now enriched by a full v1/v2 usage history. The AI Tutor answers questions and generates practice items grounded in that content, citing the source material/topic.

8.7 Classical ML Prediction (v3)

A classical ML model (e.g., gradient-boosted trees or logistic regression) trained on engagement features (session frequency, time-on-task trend) and mastery-trajectory features (stagnating/declining mastery, exam-prep accuracy trend) outputs a risk score and top contributing factors per student, reviewed each term against actual outcomes.

9. Appendices

9.1 Priority / Version Definitions

Label	Meaning
v1 — Critical/High	Required for initial commercial launch
v2	Required for the first post-launch expansion (offline + parent engagement)
v3	Planned once v1/v2 usage data exists to train classical ML models

9.2 Open Issues

- Confirm the SMS delivery provider/gateway per operating region for v2.
- Confirm minimum historical-data volume needed before v3 ML models are reliable.
- Confirm data-retention period for minor students' records against applicable national policy.

9.3 Traceability Note

Each functional requirement should be traced to specific test cases and backlog user stories, with particular attention to child-data-privacy, exam-integrity, and offline-sync-reliability coverage, before sprint planning begins.